

Automatic Grey Water Lifting Unit

AmaDrainer Box Mini

Type Series Booklet



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Type Series Booklet AmaDrainer Box Mini

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Building Services: Drainage

Lifting Units

AmaDrainer Box Mini



Main applications

Single-pump units, above-floor box Mini

- Automatic disposal of waste water from washbasins, showers, washing machines, dishwashers, etc.

Fluids handled

Standard design

- Chemically neutral grey water
- Slightly contaminated grey water ($\leq 50^\circ\text{C}$)
- Wash water (up to 90°C max. for short periods $t \leq 3$ minutes)

Variant C for aggressive fluids

- Swimming pool water
- Brackish water
- Seawater
- Water containing salt
- Aggressive fluids
- Condensate from heat recovery applications



AmaDrainer Box Mini is not suitable for sewage from urinals and toilets. Use a MiniCompacta sewage lifting unit for this purpose.

Operating data

Table 1: Operating properties

Characteristic		Value
Flow rate	Q [m^3/h]	≤ 10
	Q [l/s]	$\leq 2,8$
Head	H [m]	$\leq 6,5$
Fluid temperature	T [$^\circ\text{C}$]	$\leq +50$ ¹⁾

Design details

Design

- Collecting tank made of impact-resistant plastic
- Odour-proof cover with activated carbon filter and integrated overflow protection
- Submersible motor pump which starts and stops automatically, with float switch
- Integrated swing check valve
- To EN 12050-2

Installation

- Stationary above-floor installation
- Stationary wall mounting

Drive

- AC motor
- Integrated temperature switches
- Power cable incl. shockproof plug

Impeller type

- Open multi-vane impeller
- Free passage= 10 mm

Connections

- Inlet DN 32 / DN 40 / DN 50
- Discharge-side connection DN 40
- Shower connection DN 50
- Venting via DN 40 spigot and socket connection

¹⁾ Up to a maximum of 3 minutes for waste water from the hot water cycle of washing machines and dishwashers for non-industrial use

Designation

Example: AmaDrainer Box Mini C

Table 2: Designation key

Code	Description	
AmaDrainer Box	Type series	
Mini	Small lifting unit	
C	Material variant	
	C	Variant for aggressive fluids
	- ²⁾	Standard design

Materials

Table 3: Overview of available materials

Part No.	Component	Material
591	Collecting tank	ABS
655	Pump	See AmaDrainer 301 / 301 C type series booklet

Product benefits

- Impact-resistant plastic collecting tank
- Ready-to-connect, easy installation and commissioning
- Integrated swing check valve
- Shower connection as standard

Product information

Product information as per Regulation No. 1907/2006 (REACH)

For information as per European chemicals regulation (EC) No. 1907/2006 (REACH) see <https://www.ksb.com/en-global/company/corporate-responsibility/reach>.

Certifications

Table 4: Overview

Label	Effective in:
 Bauart geprüft und überwacht www.tuv.com ID 1111220216	Europe

Technical data

AmaDrainer Box Mini

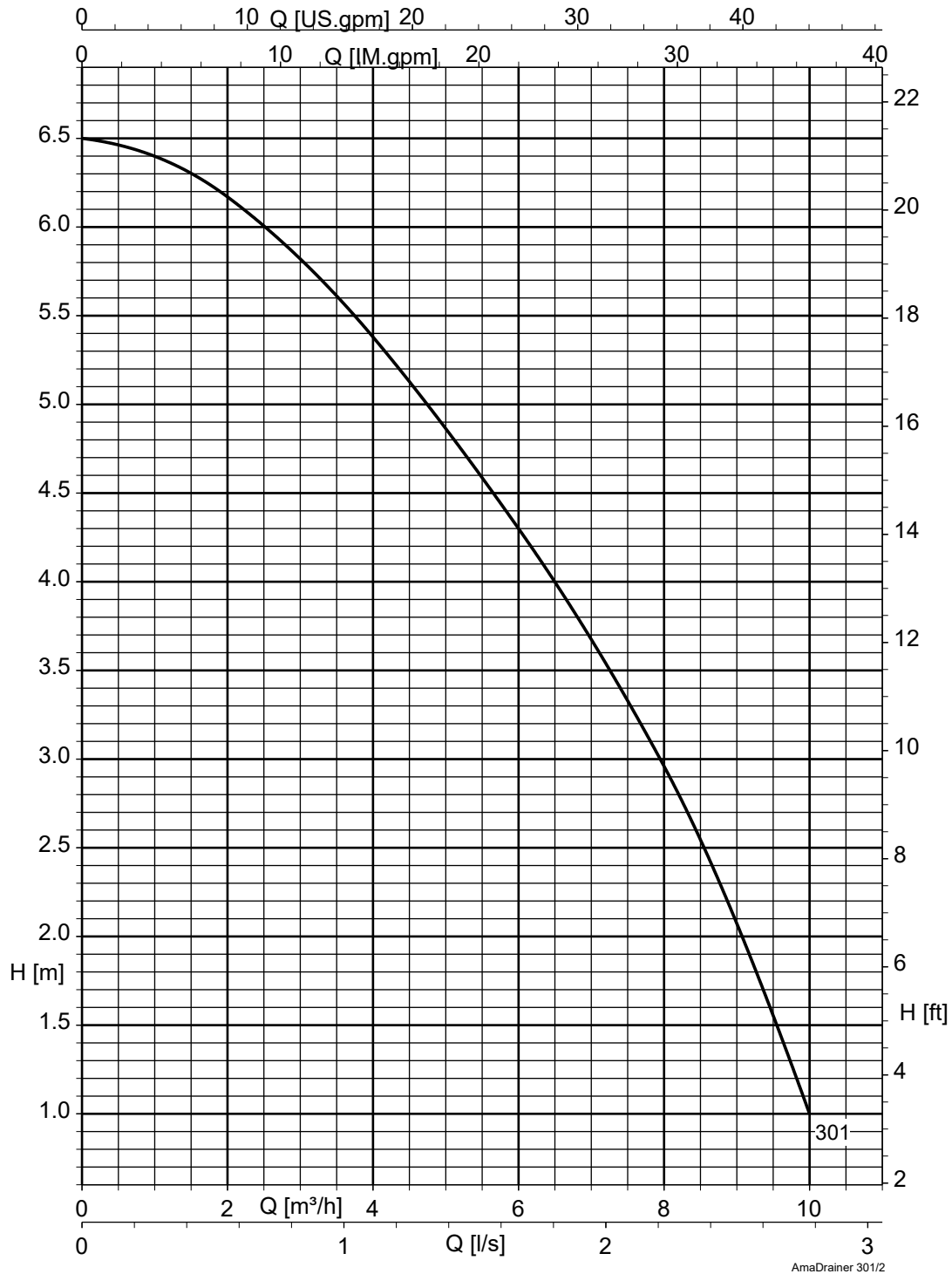
Table 5: Selection table

Design	Free passage	P ₁	P ₂	I _N	Power cable		Mat. No.	[kg]
				1~230 V	H05RN8-F			
	[mm]	[kW]	[kW]	[A]	[m]	[mm²]		
AmaDrainer Box Mini with AmaDrainer 301	10	0,43	0,25	1,9	3	3 × 0,75	29135139	9
AmaDrainer Box Mini with AmaDrainer 301 C	10	0,43	0,25	1,9	3	3 × 0,75	29135140	9

² Blank

Characteristic curve

AmaDrainer Box Mini; n = 2800 rpm



Dimensions and connections

Dimensions

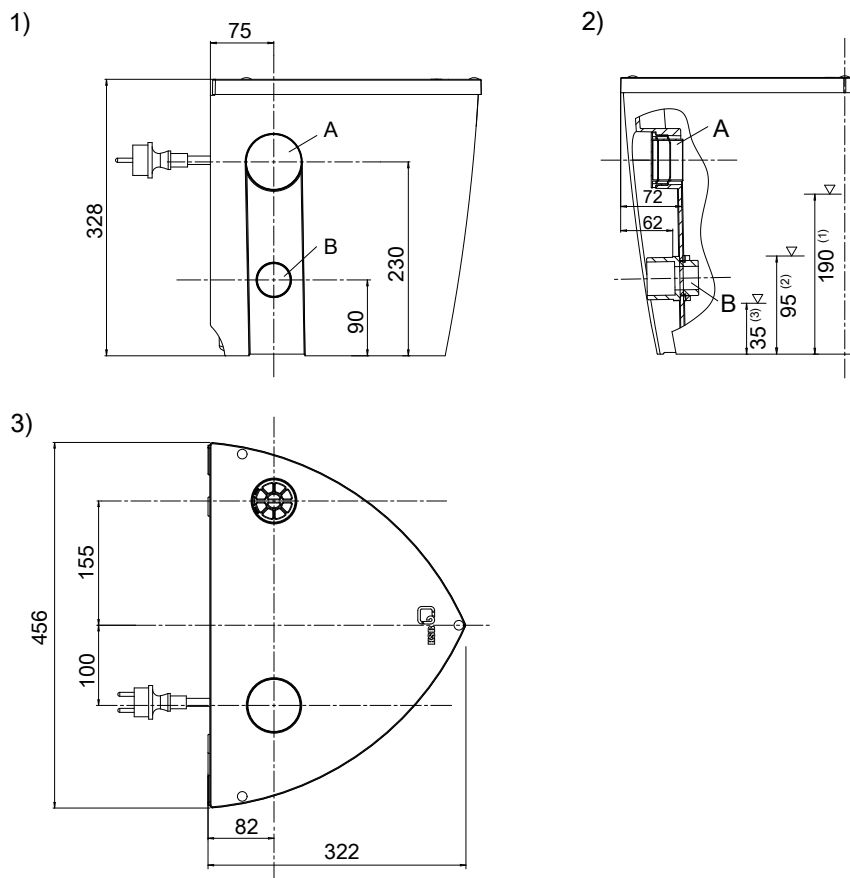


Fig. 1: Dimensions [mm]

1)	Cut-in level (standard design)
2)	Cut-in level (shower connection)
3)	Cut-out level (standard design and shower connection)
A	Inlet / discharge outlet
B	Connection to a shower

Connections

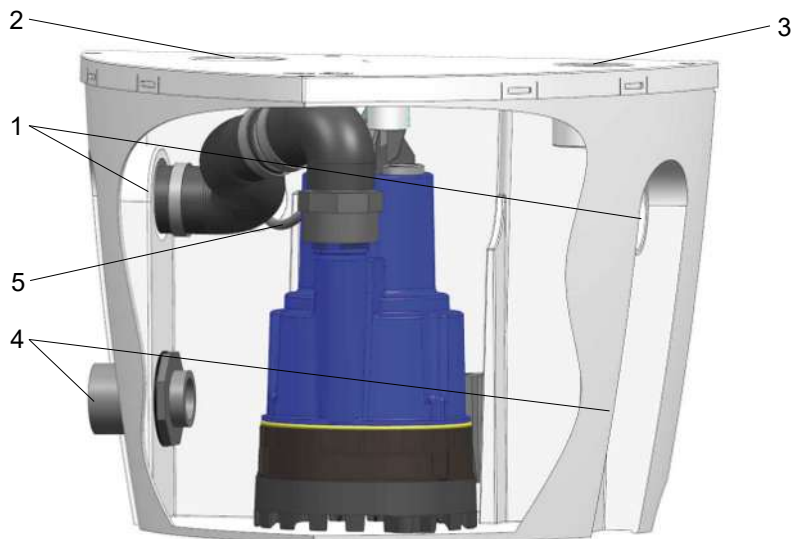


Fig. 2: Connections

Table 6: Overview of connections

Item	Connection	Description
1	Inlet line connection / discharge line connection	Socket DN 50, horizontal, on both sides for connecting an inlet line with fitting DN 32 / DN 40 / DN 50 or for connecting a discharge line with fitting DN 40
2	Inlet line connection	Socket DN 50, vertical for connecting an inlet line with fitting DN 32 / DN 40 / DN 50 or for connecting a washing machine (via a Y-drain)
3	Vent	Spigot and socket connection, DN 40 / 50, vertical for vent line (max. 40 mm outside diameter) or carbon filter insert with overflow protection
4	Shower connection	Socket, DN 50, horizontal, on both sides for connecting an inlet line (shower connection) with fitting DN 32 / DN 40 / DN 50
5	Cable entry	Cable grommet (diameter 48 mm)

Installation example

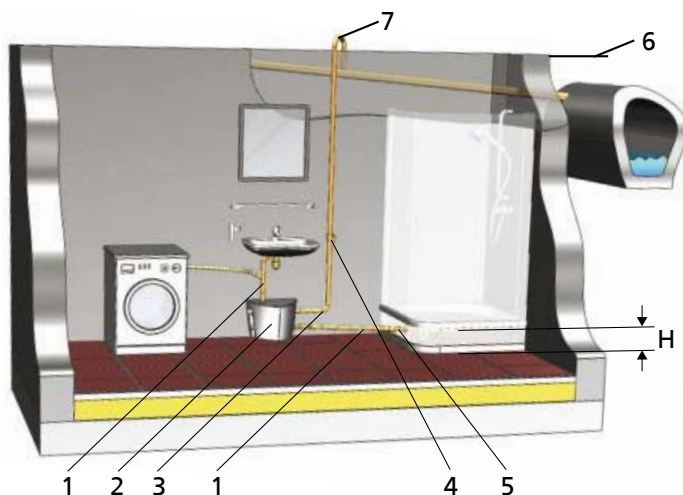


Fig. 3: Installation example

1	Inlet line	5	Valve preventing backflow into the shower basin
2	Grey water lifting unit	6	Flood level
3	Discharge line	7	Backflow loop
4	Socket gate valve	H	Height of backflow loop (150 mm minimum)

Scope of supply

The following items are included in the scope of supply:

- Collecting tank made of impact-resistant plastic
- Integrated swing check valve
- Submersible motor pump which starts and stops automatically, with float switch
- Inlet DN 32 / DN 40 / DN 50
- Discharge-side connection DN 40
- Shower connection DN 50
- Odour-proof cover with activated carbon filter and integrated overflow protection

Accessories

Pump accessories

Table 7: Pump accessories

Item	Description	Mat. No.	[kg]
P36	Replacement kit for AmaDrainer Box Mini ³⁾  For replacing an AmaDrainer N installed in AmaDrainer Box Mini by an AmaDrainer 3 Comprising: installation parts and discharge line	19066504	0,4

Alarm switchgears for pumps, non-ATEX-compliant

Table 8: AS 0 / AS 1 / AS 2 / AS 4 / AS 5 / AS W4 / AS W8

Item	Description	Mat. No.	[kg]
E50	Alarm switchgear AS 0  With circuit breaker, acoustic signalling device with 85 dB(A), green equipment-on lamp Plastic housing, IP20, H x W x D = 140 x 80 x 57 [mm]. Use float switch, F1 leakage sensor (item E64), M1 alarm contactor or signal relay of control unit as contactor.	29128401	0,5
E51	Alarm switchgear AS 2  With circuit breaker, acoustic signalling device with 85 dB(A), green equipment-on lamp, volt-free contact for hook-up to a control station Plastic housing, IP20, H x W x D = 140 x 80 x 57 [mm]. Use float switch, F1 leakage sensor (item E64) or signal relay of control unit as contactor.	29128422	0,5
E52	Alarm switchgear AS 4  With circuit breaker, acoustic signalling device with 85 dB(A), green equipment-on lamp, volt-free contact for hook-up to a control station, self-charging power supply unit for 5 hours of operation in the event of a power failure Plastic housing, IP20, H x W x D = 140 x 80 x 57 [mm]. Use float switch (E60), F1 leakage sensor (item E64) or signal relay of control unit as contactor.	29128442	0,5
E53	Alarm switchgear AS 5  Mains-independent, with self-charging power supply unit for 10 hours of operation in the event of a power failure, mains pilot LED, fault indicator light, acknowledgement button, volt-free contact for hook-up to a control station, ready for connection with 1.8 m power cable and plug. ISO housing, IP41, H x W x D = 190 x 165 x 75 [mm]. Use float switch (E60) or signal relay of control unit as contactor.	00530561	1,7
E55	Alarm switchgear AS 1  In IP30 ISO plug housing, mains-independent, with self-charging power supply unit for 5 hours of operation in the event of a power failure, acoustic signalling device 70 dB(A) with circuit breaker and integrated signal transmitter with 3-metre power cable, max. 60 °C, not suitable for steam and condensate. 1. High water alert by suspending the sensor in a (pump) sump above the pump start-up point. 2. Water alert signal at a water level of only 1 mm by placing the sensor on the floor in areas with a flooding or leakage risk, e.g. the cellar or next to the washing machine in the kitchen or bathroom.	00533740	0,9
E330	KSB washing machine interrupter  Alarm triggered, with automatic interruption of power supply to the washing machine if the water level in the collecting tank is too high 230 V AC, 50/60 Hz, max. 16 A ohm resistive load Comprising: - housing with socket and plug - buzzer - automatic re-start - M1 alarm contactor with connection cable (5 m)	01318215	0,279

³⁾ Order the replacement kit in addition to the required AmaDrainer 3.

Item	Description	Mat. No.	[kg]
E331	 AS W4 / W8 alarm switchgear Alarm triggered with automatic interruption of power supply of up to 4 or 8 washing machines if the water level in the collecting tank is too high 230 V AC, 50/60 Hz, max. 16 A resistive load per intermediate connector Comprising: - Control unit - Buzzer - Automatic re-start - M1 alarm contactor with connection cable (5 m) 4 or 8 intermediate connectors		
	KSB AS W4 - KSB washing machine interrupter for up to 4 washing machines	19075394	4
	KSB AS W8 - KSB washing machine interrupter for up to 8 washing machines	19075395	4
E332	AS1-M alarm switchgear in ISO plug housing IP30, mains-independent, with self-charging power supply unit for 5 hours' operation in the event of a power failure, acoustic signal 70 dB(A), circuit breaker, sensor connection bush and integrated monitoring for broken wires in the sensor cable Max. ambient temperature: 60° C Complete with M1 insertion-type sensor and installation accessories for high-water alert 230 V~/9 V = 1,5 VA	19074516	0,5

Control unit/switchgear accessories

Item	Description	Mat. No.	[kg]
	 M 1 alarm contactor for AmaDrainer Box Mini Magnetic float switch as contactor for alarm switchgears AS 0, AS 2, AS 4 and AS 5 or LevelControl Basic 2, with 5 m power cable	18041966	0,8
E64	 F1 leakage sensor contactor for alarm switchgears AS 0, AS 2, AS 4 or as alarm transmitter for LevelControl Basic 2 Alarm transmission options: High water alert by suspending the sensor in a (pump) sump above the pump start-up point. Warning at a water level of 1 mm in areas with a flooding or leakage risk (e.g. in the cellar or next to the washing machine in the kitchen or bathroom) Dimensions [mm]: 52 x 21 x 20 (H x W x D)	19072366	0,2
E70	 Horn, 12 V DC, 105 dB, 150 mA, IP54 Suitable for indoor installation and outdoor installation. Protect against moisture.	01086547	0,1
E71	 Alarm combination (yellow lamp and piezo buzzer 92 dB), 12 V DC, 120 mA, IP65	01139930	0,1



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